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Northern Beaches Secondary College

Manly Campus

2020 HIGHER SCHOOL CERTIFICATE

Trial Examination]

Mathematics Advanced

General Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided
- In Questions 11 – 37, show relevant mathematical reasoning and/ or calculations

**Total marks:
100**

Section I – 10 marks (pages 1-5)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 6-25)

- Attempt Questions 11 – 37
- Allow about 2 hour and 45 minutes for this section

MARKS	MC	Q11-18	Q19-24	Q25-29	Q30-33	Q34-37		TOTAL
STUDENT MARK								
MAXIMUM	10	17	18	16	19	20		100

Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 –10

Q1. Given $\log_3 5 = 1.46$ and $\log_3 2 = 0.63$, evaluate $\log_3 100$

- A. 2.09
- B. 2.72
- C. 3.55
- D. 4.18

Q2. Given $f(x) = \log_e (x - 3)$ and $g(x) = x + 4$, the domain of $f(g(x))$ is

- A. $x \geq -4$
- B. $x > -1$
- C. $x > 3$
- D. $x \geq 3$

Q3. The sample space when a fair die is rolled is $\{1, 2, 3, 4, 5, 6\}$.
Each outcome is equally likely.

For which pair of events, A and B, are A and B independent?

- A. $A = \{2, 3\}$ and $B = \{2, 3, 5, 6\}$
- B. $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$
- C. $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$
- D. $A = \{1, 3, 5\}$ and $B = \{3, 6\}$

Q4. The function $f(x)$ has derivative $f'(x) = 8\sin(4x)$ and $f\left(\frac{\pi}{4}\right) = 1$

Which of the following is $f(x)$?

- A $8\cos(4x) + 7$
- B $-2\cos(4x) - 1$
- C $2\cos(4x) + 3$
- D $2\sin(4x) + 1$

Q5. Let $f(x)$ be a differentiable function such that

- $D : (-\infty, \infty)$
- $f'(-2) = 0$; and
- $f'(x) > 0$ if $x \neq -2$

At $x = -2$, the graph of $y = f(x)$ has a

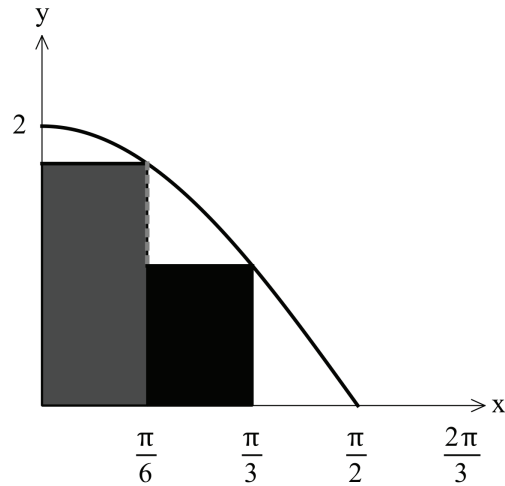
- A. horizontal point of inflection
- B. maximum turning point
- C. minimum turning point
- D. vertical asymptote

Q6. The graph of $y = f(x)$ undergoes a vertical dilation by a factor of a , followed by a horizontal translation of b units to the right, and is then reflected in the x axis.

The resulting graph is

- A. $y = -af(x-b)$
- B. $y = af(x+b)$
- C. $y = -f(ax+b)$
- D. $y = f(ax-b)$

Q7. The area under the curve $y = 2\cos x$, as shown below, is approximated by two rectangles



The value of the approximation is ?

A 1

B $\frac{\pi(\sqrt{3} + 1)}{6}$

C $\sqrt{3} + 1$

D $2\left(\frac{\pi}{6} + \frac{\pi}{3}\right)$

Q8. Which of the following is the correct value S_n for the series given below?

$$S_n = 5 + 15 + 45 + \dots + 98\,415$$

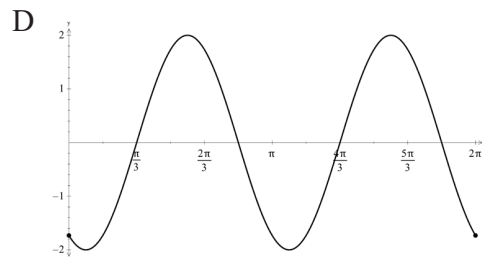
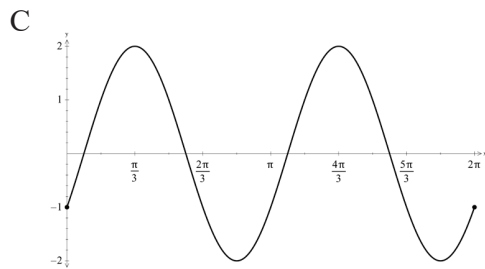
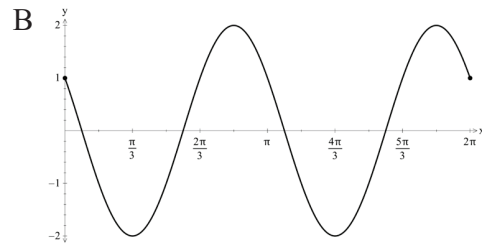
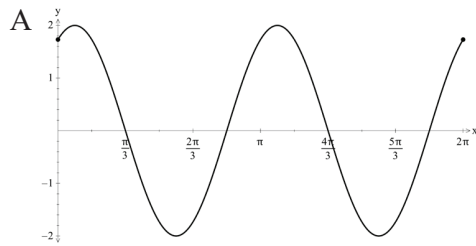
A. 125 640

B. 147 620

C. 155 600

D. 162 350

Q9. The correct graph for $y = -2\sin\left(2x + \frac{\pi}{3}\right)$ for $0 \leq x \leq 2\pi$ is



Q10. The discrete random variable X has the following probability distribution.

x	0	1	2	3	6
$P(X=x)$	$\frac{1}{4}$	$\frac{9}{20}$	$\frac{1}{10}$	$\frac{1}{20}$	$\frac{3}{20}$

Let μ be the mean of X .

$P(X < \mu)$ is?

A. $\frac{1}{4}$

B. $\frac{7}{10}$

C. $\frac{4}{5}$

D. $\frac{17}{20}$

End of Multiple Choice

Write your student exam number in the boxes



Northern Beaches Secondary College

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Mathematics Advanced

Trial Examination

Section II – Answer Booklet

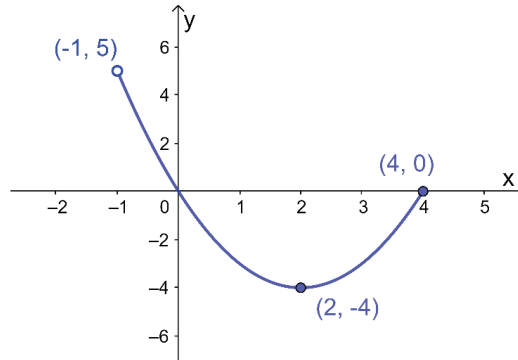
90 marks

Attempt Questions 11 – 37

Allow about 2 hour and 45 minutes for this section

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response. Additional Writing space is available at the end of the booklet.
- Your responses should include relevant mathematical reasoning and/or calculations.

Q11. The graph of a function $f(x)$ is shown below



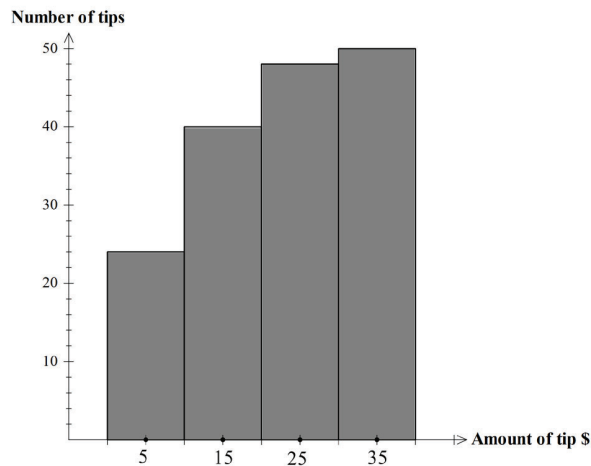
Using interval notation, state the domain and range of $f(x)$

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Q12. Peter works in a restaurant, and each customer can leave him a tip, some extra money, for good service. On one day, Peter earns 50 tips, as shown in the cumulative histogram:



One of the tips was incorrectly listed as \$15 instead of \$25.

Giving reasons, explain what effect this would have on each of the following statistics:

i. The median:

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ii. The mean:

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- Q13. Sarita opens an annuity account, which requires contributions to be made at the end of each period. The table below shows the future of an annuity of \$1 for various interest rates per period.

Future value of an annuity of \$1

Periods	Interest rate per period			
	0.25%	0.75%	1.5%	3%
2	2.0025	2.0075	2.015	2.03
6	6.0376	6.1136	6.2296	6.4684
12	12.1664	12.5076	13.0412	14.1920
24	24.7028	26.1885	28.6335	34.4265

To save for a computer for university, Sarita makes regular deposits at the end of each month into an annuity account that pays 3% p.a. The annuity account compounds interest monthly. How much will Sarita have to deposit each month if she expects the computer to cost \$2800 at the end of two years?

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- Q14. Differentiate $e^{x \sin x}$.

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- Q15. If $f(x) = \log_e(x^2 + 1)$, calculate $f'(-1)$.

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Q16. A shop assistant is building a display of soup cans. The top row will contain 3 cans, and each subsequent row below will have 2 more cans than the previous row above.

- i. How many cans would be in the 10th row?

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- ii. The shop assistant has 400 cans. How many rows is the tallest display that he can build using this pattern?

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Q17. The completion times for the Oztown triathlon race were normally distributed with mean time 60 minutes and standard deviation 5 minutes. Using the empirical rule, find Ozzie's completion time if he finished ahead of 84% of competitors.

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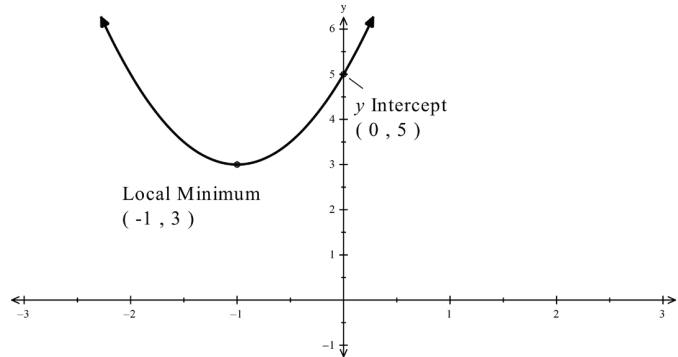
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Q18. The function $f(x) = (x - 1)^2$ is transformed and the equation of the new function is of the form $y = k f(x + a) + c$ where k , a and c are constants.

The graph of the new function is shown below.



What are the values of k , a and c ?

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Q19. Two events, A and B , are such that $P(A) = \frac{3}{5}$ and $P(B) = \frac{1}{4}$.

If A' denotes the complement of A , calculate $P(A' \cap B)$ when

i) $P(A \cup B) = \frac{3}{4}$

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ii) A and B are mutually exclusive.

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Q20. Given the function $g(x) = \frac{3}{2x-1} + 1$

- i. What is the equation of the vertical asymptote? 1

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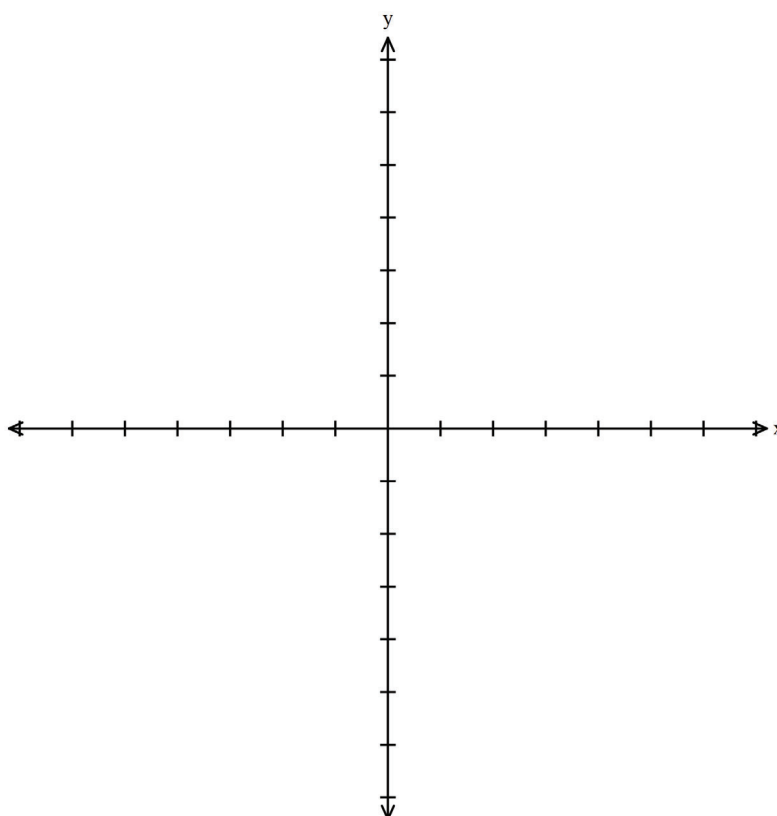
- ii. Sketch the function $g(x)$ in the space below, labelling asymptotes and intercepts. 2

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- iii. The function $g(x)$ was derived from applying transformations to the function $f(x) = \frac{1}{x}$.

What was the horizontal translation applied to achieve this transformation? 1

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Q21. Find

i. $\int 24(2x - 7)^5 dx$

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ii. $\int \frac{2x + 2}{4x^2 + 8x + 1} dx$

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Q22. Given the function $f(x) = x^2 - 1$ and $g(x) = \sqrt{4 - x^2}$,

sketch $y = f(g(x))$ over its natural domain.

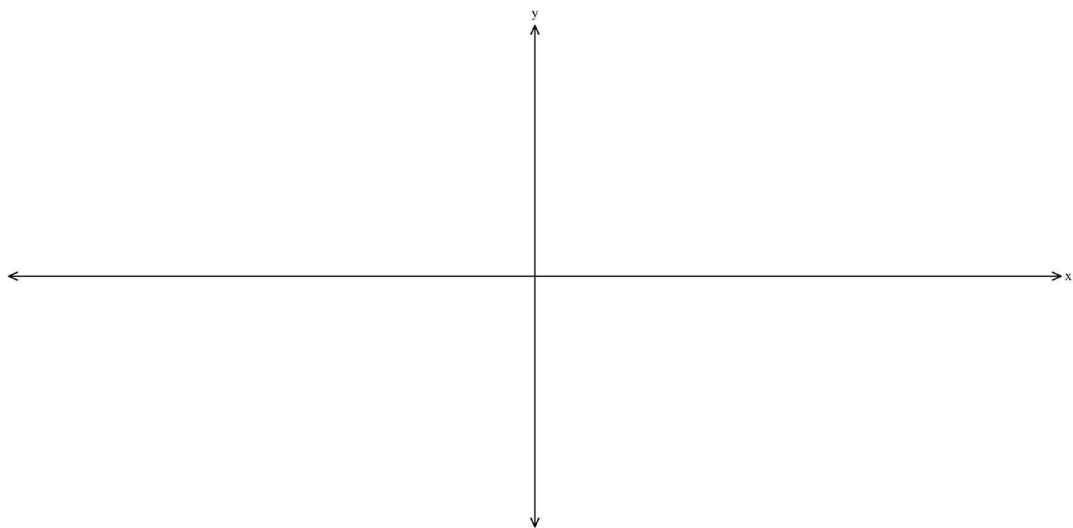
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Q23. Given the series $\tan^2 x - \tan^4 x + \tan^6 x - \tan^8 x + \dots$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

- i) Determine the values of x such that the series has a limiting sum. 2

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- ii) Write a fully simplified expression in x for the limiting sum of the given series. 1

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Q24. Solve $2\cos(3\theta) = -1$ for $0 \leq \theta \leq \pi$ 3

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Q25. i. Differentiate $\sin^2 x$ 1

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ii. Hence calculate $\int_0^{\frac{\pi}{4}} (\sin x + \cos x)^2 dx$, (Leave your answer in exact form.) 3

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Q26. A six-sided die is biased as shown in the probability distribution below.

$X = x$	1	2	3	4	5	6
$P(X = x)$	0.1	0.25	0.05	0.3	0.17	0.13

i. Find the expected value of X . 1

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ii. Hence, using part i, calculate the variance. 2

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Q27. The body weights and metabolic rate recordings for five women are shown below.

Weight (kg)	52	63	65	47	49
Metabolic rate	1671	1669	1812	1442	1607

- i. Find the correlation coefficient, r . (correct to 3 decimal places) 1

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- ii. Find the coefficients of the linear regression line, $y = mx + c$, where x is the weight and y is the metabolic rate. 1

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- iii. Verify that $m = r \frac{s_y}{s_x}$, where s_x and s_y are the standard deviations of the x and y values respectively. 2

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Questions 11 to 27 are worth 46 marks in total
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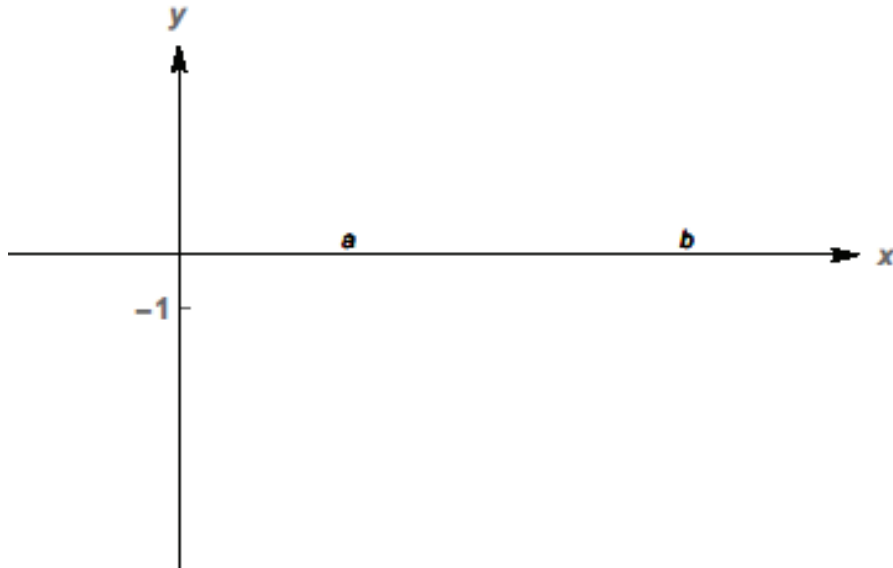
Question 28 start on next page.

Q28. Over the domain $[a, b]$, function f satisfies the following conditions:

$$f(x) < -1, f'(x) < 0 \text{ and } f''(x) > 0.$$

- i. Sketch a possible graph of $y = f(x)$ for the domain $[a, b]$.

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- ii. Write an expression for the minimum value of the function $g(x) = 1 - f(x)$ over the domain $[a, b]$.

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Q29. A particle is moving along the x axis. Its displacement, x metres from the origin after t seconds, is given by $x = e^t - 3e^{2t}$ for $t \geq 0$.

Explain why the particle never comes to rest.

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Q30. Given the function $f(x) = -x^3 + 9x^2 - 24x + 16$

- i. Determine the coordinates of any stationary points and their nature. 3

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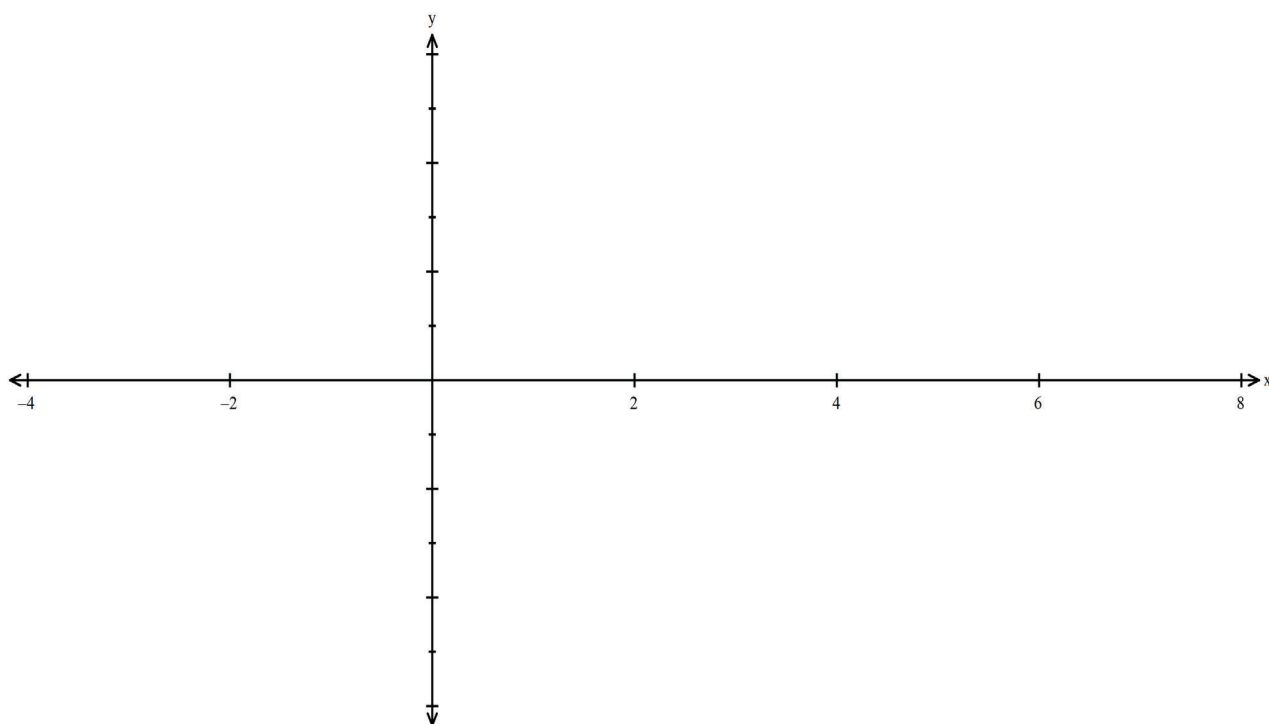
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- ii. Labelling the stationary points and intercepts with the coordinate axes, sketch the graph of $y = f(x)$ over the domain $[0, 6]$. 3



Question 30 continues on the next page.

Question 30 continued

- iii. State the global minimum of $y = f(x)$ over the domain. $[0, 6]$

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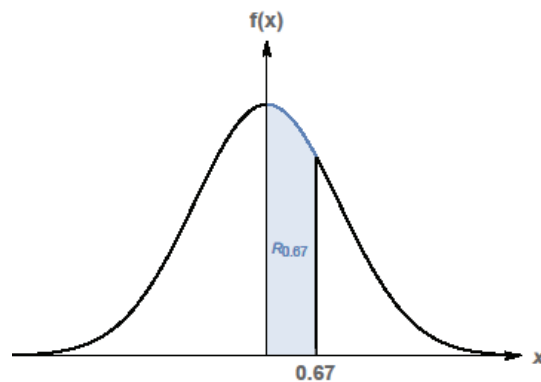
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Q31. The random variable X has the standard normal distribution, with probability density

function $f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$ over the domain $(-\infty, \infty)$.

In the diagram below, the region $R_{0.67}$ indicates the area under $y = f(x)$ over the domain $[0, 0.67]$.



If the area of $R_{0.67}$ is 0.25, determine the range of scores that could be identified as outliers.

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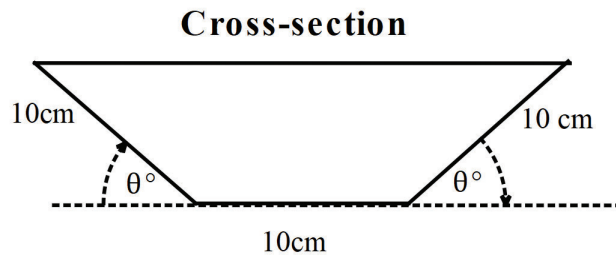
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Q32. The diagram below shows a rectangular piece of metal, which is bent to form a gutter for the flow of water. The cross-section of a gutter is a trapezium.



- i. Show that the cross-sectional area $A(\theta)$, as a function of θ , is given by 2

$$A(\theta) = 100\sin\theta(1 + \cos\theta)$$

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- ii. Find the angle θ that maximizes the cross-sectional area. 3

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Q33. i. Prove that $\frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} \equiv 2\sec^2 x$ where $\sin x \neq \pm 1$

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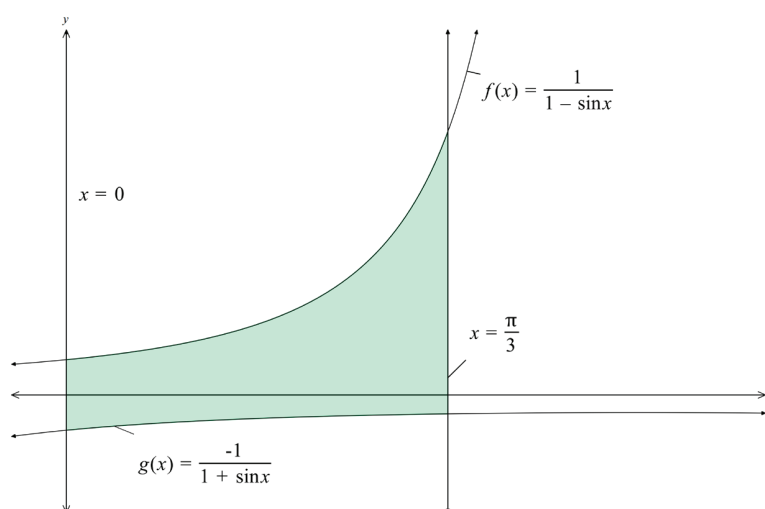
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ii. The region shown below represents the area between the curves

$$f(x) = \frac{1}{1 - \sin x} \text{ and } g(x) = \frac{-1}{1 + \sin x}, \text{ for the domain } \left[0, \frac{\pi}{3}\right].$$



Calculate the area between the 2 curves for the given domain.

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Q34. The lizards on Goanna Island are gradually dying out. The predicted population P of lizards, t years after the first observation was made, is $P = P_0 e^{-0.01t}$, where P_0 is the initial population.

- i. At what rate is the population changing 45 years later? Answer in terms of P_0 . 1

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- ii. What percentage of the original population remains 45 years later, correct to the nearest 1%? 1

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- iii. When will the population decline to 10% of the original population? Leave your answer in exact form. 2

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Q35. The monthly sales, X (in units of \$100, 000), of a business are modelled by the two probability density functions:

$$g(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{Model 1}$$

OR

$$f(x) = \begin{cases} kx^3(1-x^2) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{Model 2}$$

- i. Determine the value of k in the second model. 2

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- ii. Show that the two models have the same median value. 3

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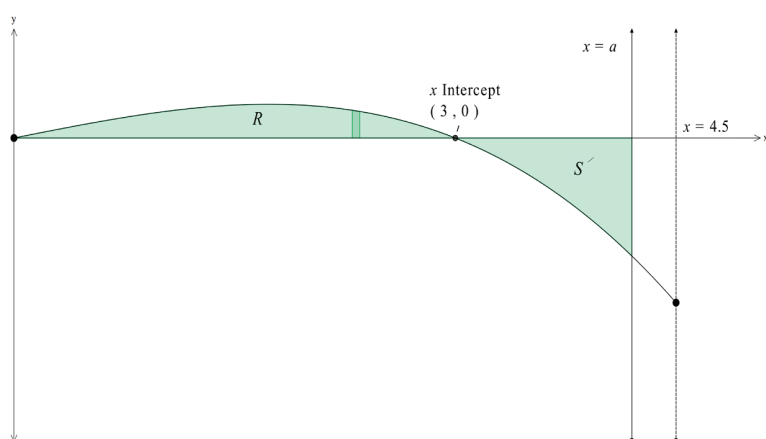
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Q36. The graph below shows the cubic curve $y = 9x - x^3$, with domain $[0, 4.5]$.



The curve crosses the x -axis at $x = 0$ and $x = 3$.

The diagram shows two bounded regions:

R bounded by $y = f(x)$, $y = 0$, $x = 0$ and $x = 3$

S bounded by $y = f(x)$, $y = 0$, $x = 3$ and $x = a$

It is given that $a > 3$. Determine the value of a such that regions R and S have the same area.

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Q37. Sam invests $\$M$ into an account at the beginning of each month. The account earns 3% p.a. interest, compounded monthly.

- i. Show that the total amount in the account at the end of the second month, A_2 is given by $A_2 = M \times (1.0025 + 1.0025^2)$ 1

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- ii. Show that the amount in the account after n months, A_n , is given by $A_n = 401M(1.0025^n - 1)$ 2

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Question 37 continues on page 25

Question 37 continued

- iii. Hence, calculate how much is needed to invest at the beginning of each month if the plan is to have \$6000 at the end of 12 months.

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- iv. After the first five months of contributions, Sam increases his monthly payment to \$600. Given Sam's final contribution is less than \$600, what will be the minimum amount of this contribution (to the nearest \$5) to reach his goal of \$6000?

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End of Paper.

HSC Mathematics Advanced 2020 Trial Exam Solutions

Q1-D Q2 - B Q3-D Q4-B Q5-A
 Q6- A Q7- B Q8 – B Q9 – D Q10 - B

Q1	$\begin{aligned} & \log_3 100 \\ &= 2\log_3 10 \\ &= 2\log_3 (2 \times 5) \\ &= 2(\log_3 5 + \log_3 2) \\ &= 2(1.46 \times 0.63) \\ &= 4.18 \end{aligned}$	D
Q2	Domain $f(x)$ - $x > 3$ Domain $g(x)$ - all real x $\begin{aligned} g(x) &= x + 4 \\ f(x) &= \log_e (x - 3) \end{aligned}$ $\begin{aligned} f(g(x)) &= \log_e [(x + 4) - 3] \\ &= \log_e (x + 1) \end{aligned}$ Domain $x > -1$	B
Q3	A and B independent not to be confused with mutually exclusive. $P(A \text{ and } B) = P(A) \times P(B)$ $\begin{aligned} A : \quad & \frac{2}{6} \neq \frac{2}{6} \times \frac{4}{6} \\ B : \quad & 0 \neq \frac{1}{2} \times \frac{1}{2} \\ C : \quad & \frac{1}{6} \neq \frac{1}{2} \times \frac{1}{2} \\ D : \quad & \frac{1}{6} = \frac{3}{6} \times \frac{2}{6} \end{aligned}$	D

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Q4	$f(x) = \int 8\sin(4x) \, dx$ $= -2 \int -4\sin(4x) \, dx$ $f(x) = -2\cos(4x) + C$ $1 = -1\cos\left(4 \times \frac{\pi}{4}\right) + C$ $1 = -1 \times -1 + C$ $C = -1$ $f(x) = -2\cos(4x) - 1$	B
Q5	Single stationary point where all else is increasing therefore point of horizontal inflection	A
Q6	$y = af(x)$ $y = af(x - b)$ $y = -af(x - b)$	A
Q7	$A = \frac{\pi}{6} \times 2\cos\frac{\pi}{6} + \frac{\pi}{6} \times 2\cos\frac{\pi}{3}$ $= \frac{2\pi}{6} \left(\frac{1}{2} + \frac{\sqrt{3}}{2} \right)$ $= \frac{\pi(1 + \sqrt{3})}{6}$	B
Q8	$a = 5 \quad r = 3$ $98415 = 5 \times 3^{n-1}$ $19683 = 3^{n-1}$ $\log 19683 = (n-1) \log 3$ $n-1 = 9$ $n = 10$ $S_{10} = \frac{5(3^{10} - 1)}{3 - 1} = 147620$	B

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Q9	$y = -2 \sin \left[2 \left(x + \frac{\pi}{6} \right) \right]$	D
Q10	Mean $= E(X) = 0 \times \frac{1}{4} + 1 \times \frac{9}{20} + 2 \times \frac{1}{10} + 3 \times \frac{1}{20} + 6 \times \frac{3}{20}$ $= 1.7$ $\therefore$ $P(X = 0 \text{ or } 1) = \frac{1}{4} + \frac{9}{20} = \frac{14}{20} = \frac{7}{10}$	B

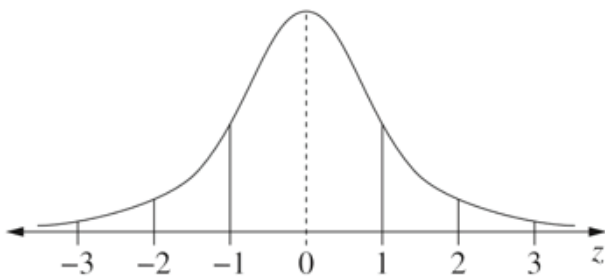
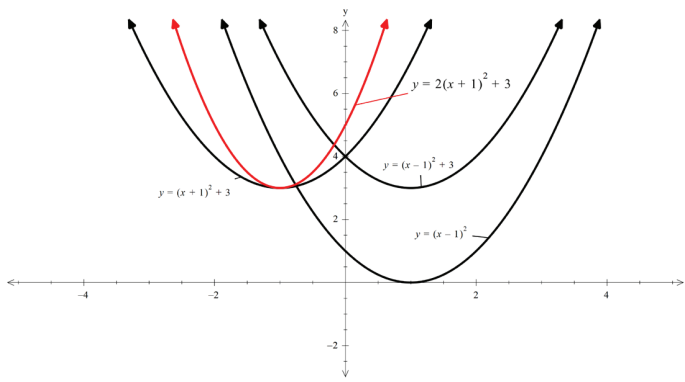
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		Mark
Q11	Domain is $(-1, 4]$ Range is $[-4, 5)$	2 marks – both correct 1 mark – one correct
Q12a	Sample Answer: Through use of the OGIVE, it can be seen that the median is lower than 15 therefore the error has no effect on the median.	1 mark – correct answer with correct explanation
Q12b	Sample Answer: Since the mean calculation uses the specific value of each tip, increasing a single tip from \$15 to \$25 must increase the mean by $\frac{\\$10}{50} = \\0.20.	1 mark – correct answer with correct explanation
Q13	$3\% \text{ pa} = 0.25\% \text{ pm}$ 2 years = 24 months $\\$ \frac{2800}{24.7028} = \\$113.347 \approx \\$113.35$	2 marks – correct answer 1 mark – correct monthly interest rate.

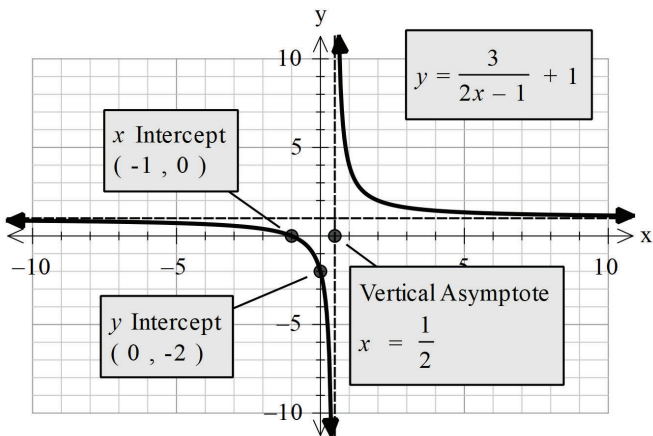
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Q14	$y = e^{x \sin x}$ $f(x) = x \sin x$ $f'(x) = \sin x + x \cos x$ $y = e^{f(x)}$ $\frac{dy}{dx} = f'(x) e^{f(x)}$ $= (\sin x + x \cos x) e^{x \sin x}$	2 correct answer 1 mark - Correct derivative for index - Incorrect index derivative used correctly
Q15	$f(x) = \log_e (x^2 + 1)$ $f'(x) = \frac{2x}{x^2 + 1}$ $f'(-1) = -\frac{2}{2} = -1$	2 marks – correct answer 1 mark – correct derivative. - Correct substitution
Q16-i	Arithmetic Series $a = 3 \quad d = 2$ $T_{10} = a + (n - 1)d$ $= 3 + 9 \times 2 = 21$	1 mark – correct answer
Q16-ii	$Sn = 400$ $Sn = \frac{n}{2}[2a + (n - 1)d]$ $400 = \frac{n}{2}[6 + 2n - 2]$ $800 = 4n + 2n^2$ $0 = n^2 + 2n - 400$ $n = \frac{-2 \pm \sqrt{4 + 4 \times 400}}{2}$ $= (-1 \pm \sqrt{401})$ $= 19.02 \text{ or } -21.02$ $\therefore 19 \text{ rows high}$	2 – correct answer 1 mark – correct substitution into formula.

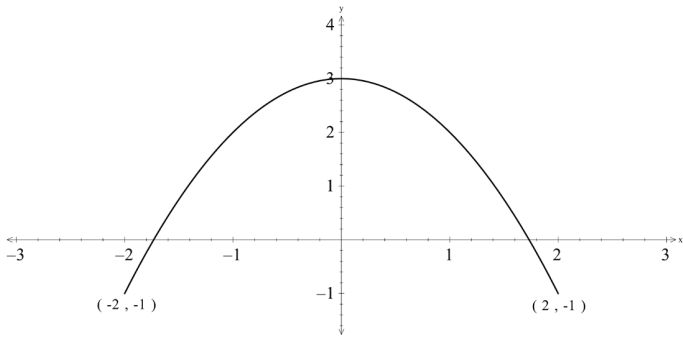
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Q17	Normal distribution  • approximately 68% of scores have z-scores between -1 and 1 • approximately 95% of scores have z-scores between -2 and 2 • approximately 99.7% of scores have z-scores between -3 and 3 Therefore must be greater than one standard deviation less than mean to be finish ahead of 84% Therefore less than 55 minutes.	2 – correct answer 1 mark – correctly identifies needs to be one SD from mean.
Q18	Vertex moved up 3 $\therefore c = 3$ and to left 2 places $y = k(x + 1)^2 + 3 \quad y = f(x) = (x - 1)^2$ $x = 0 \quad y = 5 \quad y = f(x + 2) = \{(x + 2) - 1\}^2$ $5 = k \times 1^2 + 3 \quad y = (x + 1)^2$ $k = 2$ $k = 2 \quad a = 2 \quad c = 3$ 	2 marks – all three letters correct value 1 mark – 2 correct
Q19i	$P(A' \cap B) = \frac{3}{20}$	1 correct solution
Q19ii	$P(A' \cap B) = \frac{5}{20}$	1 correct solution

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Q20i	$x = \frac{1}{2}$	1 mark correct solution
Q20ii		2 marks correct solution 1 mark some missing components
Q20iii	Right $\frac{1}{2}$ unit (if follows dilation); Right 1 unit if precedes dilation,	1 mark correct solution
Q21i	$\int 24(2x - 7)^5 dx$ $= 24 \int (2x - 7)^5 dx$ $= 24 \times \frac{(2x - 7)^6}{2 \times 6} + C$ $= 2(2x - 7)^6 + C$	1 mark correct solution
Q21ii	$\int \frac{2x + 2}{4x^2 + 8x + 1} dx$ let $y = 4x^2 + 8x + 1$ $\frac{dy}{dx} = 8x + 8 = 4(2x + 2)$ $\frac{1}{4} \int \frac{4(2x + 2)}{4x^2 + 8x + 1} dx$ $= \frac{1}{4} \ln 4x^2 + 8x + 1 + C$	2 marks correct solution 1 mark incorrect multiple of correct log function

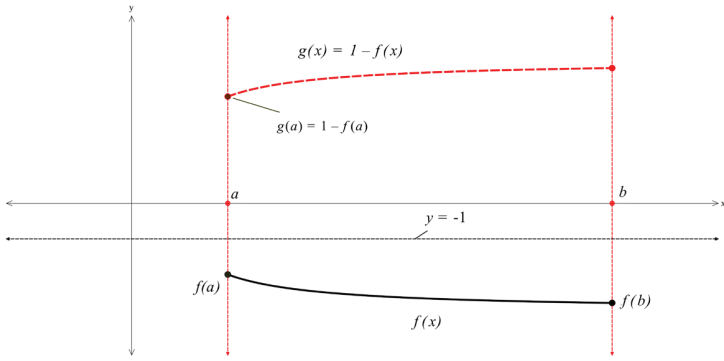
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Q22		3 marks correct solution 2 marks one missing feature 1 mark two missing features
Q23i	$r = \frac{-\tan^4 x}{\tan^2 x}$ $= -\tan^2 x$ $\therefore -\tan^2 x < 1$ $-\frac{\pi}{4} < x < \frac{\pi}{4}$	2 marks correct solution 1 mark correct inequality for limiting sum
Q23ii	$S_{\infty} = \frac{\tan^2 x}{1 - (-\tan^2 x)}$ $= \frac{\tan^2 x}{\sec^2 x}$ $= \sin^2 x$	1 mark correct solution
Q24	$2\cos(3\theta) = -1$ if $0 \leq \theta \leq \pi$ then $0 \leq 3\theta < 3\pi$ $\cos 3\theta = -\frac{1}{2}$ $3\theta = \cos^{-1}\left(-\frac{1}{2}\right)$ $3\theta = \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{8\pi}{3}$ $\theta = \frac{2\pi}{9}, \frac{4\pi}{9}, \frac{8\pi}{9}$	3 marks correct solution 2 marks only one error in correct process 1 mark correct ref angle and quads for solution.

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Q25-i	$y = \sin^2 x$ $\frac{dy}{dx} = 2 \cos x \sin x$	1 mark correct solution
Q25-ii	$\int_0^{\frac{\pi}{4}} (\sin x + \cos x)^2 dx$ $= \int_0^{\frac{\pi}{4}} \sin^2 x + 2 \sin x \cos x + \cos^2 x dx$ $= \int_0^{\frac{\pi}{4}} 1 + 2 \sin x \cos x dx$ $= \left[x + \sin^2 x \right]_0^{\frac{\pi}{4}}$ $= \left(\frac{\pi}{4} + \left(\frac{1}{\sqrt{2}} \right)^2 \right) - 0$ $= \frac{\pi}{4} + \frac{1}{2}$	3 marks correct solution 2 marks correct definite integral 1 mark correct expansion and simplified to integral $\int_0^{\frac{\pi}{4}} 1 + 2 \sin x \cos x dx$ must get to line 3 to earn 1 mark
Q26-i	$E(x) = 1 \times 0.1 + 2 \times 0.25 + 3 \times 0.05 + 4 \times 0.3 + 5 \times 0.17 + 6 \times 0.13$ $= 3.58$	1 mark correct solution
Q26-ii	$Var(X) = E(X^2) - \mu^2$ $= 1^2 \times 0.1 + 2^2 \times 0.25 + 3^2 \times 0.05 + 4^2 \times 0.3 + 5^2 \times 0.17 + 6^2 \times 0.13 - 3.58^2$ $= 2.4636$	2 marks correct solution showing working 1 mark correct value with some relevant working
Q27-i	0.8233 =0.823	1 mark correct solution
Q27-ii	$m = 13.346$ $c = 903.499$	1 mark BOTH m and c values correct
Q27-iii	$\begin{bmatrix} s_y = 119.732 \\ s_x = 7.386 \end{bmatrix} \Rightarrow r \times \frac{s_y}{s_x} = 0.8233 \times 16.2096 = 13.346 = m$	2 marks correct solution, verified as shown 1 mark for finding $\frac{s_y}{s_x} = 16.21$ but not fully verified

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Q28-ia		2 marks for correct graph of $f(x)$. Must be below -1 and only from a to b inclusive 1 mark if correct concavity or a decreasing graph between a and b
Q28-ii	Minimum $\Rightarrow g(a)=1-f(a)$	1 correct solution
Q29	$x = e^t - 3e^{2t}$ $\frac{dx}{dt} = e^t - 6e^{2t}$ $\frac{dx}{dt} = v$ To stop $\Rightarrow v = 0$ $e^t - 6e^{2t} = 0$ $e^t(1 - 6e^t) = 0$ $e^t \neq 0 \Rightarrow t = \ln 0 \text{ not possible}$ $1 = 6e^t$ $e^t = \frac{1}{6}$ $t = \ln\left(\frac{1}{6}\right) = -1.79$ as $t \geq 0$ there are no values for $v = 0$	2 marks correct solution with reasons why velocity cannot =0 1 mark for correct derivative and putting = 0 and attempt to solve 0 marks just for stating exponentials can't = 0 as question says EXPLAIN

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Q30-i	$f(x) = -x^3 + 9x^2 - 24x + 16$ $f'(x) = -3x^2 + 18x - 24$ $= -3(x^2 - 6x + 8)$ $= -3(x - 4)(x - 2)$ $f''(x) = -3(2x - 6)$ $= -6(x - 3)$ $f'(x) = 0$ at $x = 4$ and $x = 2$ at $x = 4$ $f''(x) = -6 < 0$ concave down $\therefore$ max at $(4, 0)$ at $x = 2$ $f''(x) = 6 > 0$ concave up $\therefore$ min at $(2, -4)$	3 marks 1 mark for finding the x -coordinates of the stationary points 1 mark for the finding the y -coordinates of the stationary points (Several students did this in (ii)) 1 mark for correctly determining the nature of the stationary points
Q30-ii	Graph of the function $f(x) = -x^3 + 9x^2 - 24x + 16$. The graph shows the following key points: - y Intercept: $(0, 16)$ - x Intercept: $(1, 0)$ - Local Minimum: $(2, -4)$ - Local Maximum: $(4, 0)$ - Endpoint: $(6, -20)$	3 marks 1 mark for shape 1 mark for labelling the stationary points found in (i) 1 mark for coordinates of the endpoints (No marks for the x -intercept at 1)
Q30-iii	-20 Note the global minimum / minimum value is a single number, not a point.	1
Q31	Since $R_{0.67}$ has area of 0.25, $Q_3 = 0.67$ and $Q_1 = -0.67$ by symmetry. Hence, the outliers will be below $-2.68 = -0.67 - 1.5 \times (0.67 - (-0.67))$ or above $2.68 = 0.67 + 1.5 \times (0.67 - (-0.67))$	3 marks 1 mark for $Q_3 = 0.67$ 1 mark for IQR 1 mark for using the outlier definition

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Q32-i	Cross-sectional area $= \frac{h}{2}(a+b)$, where $h = 10 \sin \theta$, $a = 10$, $b = 10 + 2 \times 10 \cos \theta$ $A(\theta) = \frac{10 \sin \theta}{2}(10 + (10 + 20 \cos \theta))$ $= 5 \sin \theta (20 + 20 \cos \theta)$ $A(\theta) = 100 \sin \theta (1 + \cos \theta)$	2 marks 1 mark for some progress, e.g. a correct expression for the area of the rectangle
Q32-ii	$A'(\theta) = 100 [\cos \theta (1 + \cos \theta) + \sin \theta (-\sin \theta)]$ $= 100 (\cos \theta + \cos^2 \theta - (1 - \cos^2 \theta))$ $= 100 (2 \cos^2 \theta + \cos \theta - 1)$ $= 100 (\cos \theta + 1)(2 \cos \theta - 1)$ $\therefore A'(\theta) = 0$ if $\cos \theta = -1$ or $\cos \theta = \frac{1}{2}$ Since θ is acute, $\cos \theta = \frac{1}{2} \Rightarrow \theta = \frac{\pi}{3}$ $A''(\theta) = 100 (-4 \cos \theta \sin \theta - \sin \theta) = -100 \sin \theta (4 \cos \theta + 1)$ $A''\left(\frac{\pi}{3}\right) = -150\sqrt{3} < 0$ Hence, the cross-sectional area is a maximum when $\theta = \frac{\pi}{3}$	3 marks 1 mark for differentiating 1 mark for solving $A' = 0$ 1 mark for showing a maximum

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Q33-i	$ \begin{aligned} LHS &= \frac{1}{1 - \sin x} + \frac{1}{1 + \sin x} \\ &= \frac{(1 + \sin x) + (1 - \sin x)}{1 - \sin^2 x} \\ &= \frac{2}{\cos^2 x} \\ &= 2 \sec^2 x \\ &= RHS \end{aligned} $	1 mark
Q33-ii	$ \begin{aligned} &\int_0^{\frac{\pi}{3}} \left(\frac{1}{1 - \sin x} - \frac{-1}{1 + \sin x} \right) dx \\ &= \int_0^{\frac{\pi}{3}} 2 \sec^2 x \, dx \\ &= 2 \left[\tan x \right]_0^{\frac{\pi}{3}} \\ &= 2 \left(\tan \frac{\pi}{3} - 0 \right) \\ &= 2\sqrt{3} \end{aligned} $	3 marks 1 st mark for a correct expression for the area, or for some understanding of the area between two curves 2 nd mark for simplifying using (i)
Q34-i	$ \begin{aligned} \frac{dP}{dt} &= kP \\ \text{at } t &= 45 \\ P &= P_0 e^{-0.45} \\ \frac{dP}{dt} &= -0.1 P_0 e^{-0.45} \end{aligned} $	1 mark for correct solution
Q34-ii	$ \begin{aligned} \text{at } t &= 45 \\ P &= P_0 e^{-0.45} \\ \frac{P_0 e^{-0.45}}{P_0} \times 100 &= 63.76 \cong 64\% \end{aligned} $	1 mark for correct solution

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Q34-iii	$P = P_0 e^{-0.01t}$ 10% $\frac{P_0}{10} = P_0 e^{-0.01t}$ $0.1 = e^{-0.01t}$ $\ln(0.1) = -0.01t$ $t = \frac{-\ln 10}{-0.01}$ $= 100 \ln 10$	2 marks for correct working and solution 1 mark for writing the equation, then not making any progress.
Q35-i	$\int_{-\infty}^{\infty} f(x) dx = 1 \Rightarrow k \int_0^1 (x^3 - x^5) dx = 1$ Since $\int_0^1 (x^3 - x^5) dx = \left[\frac{x^4}{4} - \frac{x^6}{6} \right]_0^1 = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$ then $\frac{k}{12} = 1 \Rightarrow k = 12$	2 marks for working and solution 1 mark for integrating.
Q35-ii	Let $x = M$ be the median of model g. Then: $\frac{1}{2} = \int_0^M g(x) dx = \int_0^M 2x dx = \left[x^2 \right]_0^M = M^2, \text{ i.e. } M^2 = \frac{1}{2}$ Consider $\int_0^M f(x) dx = \int_0^M 12(x^3 - x^5) dx = \left[3x^4 - 2x^6 \right]_0^M = 3M^4 - 2M^6$ Since $M^2 = \frac{1}{2}$, $\int_0^M f(x) dx = 3\left(\frac{1}{2}\right)^2 - 2\left(\frac{1}{2}\right)^3 = \frac{1}{2}$ Hence $x = M$ is also the median of model f, i.e. the two models have the same median.	3 marks for correct solution 2 marks for finding M^2 and integrating model 2, but did not sub M. 1 mark for only finding M^2
Q36	$\int_0^3 9x - x^3 dx + \int_3^a 9x - x^3 dx = 0$ $\left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^3 + \left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_3^a = 0$ $\left(\frac{81}{2} - \frac{81}{4} \right) + \left(\frac{9a^2}{2} - \frac{a^4}{4} \right) - \left(\frac{81}{2} - \frac{81}{4} \right) = 0$ $\frac{9a^2}{2} = \frac{a^4}{4}$ $18a^2 = a^4, \quad a \neq 0$ $a^2 = 18$ $a = 3\sqrt{2} \text{ as } a > 0$	3 marks for correct solution 2 marks for integrating with different limits and then equating. However did not make progress. 1 mark for integrating with different limits.

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	Alternatively $\int_0^a 9x - x^3 \, dx = 0$ $\left[\frac{9x^2}{2} - \frac{x^4}{4} \right]_0^a = 0$ $18a^2 - a^4 = 0$ $a^2(18 - a^2) = 0$ $a = 0 \text{ or } a = \pm 3\sqrt{2}$ but $a > 0$ therefore $a = 3\sqrt{2}$	
Q37 – i	$A_1 = M \times \left(1 + \frac{3}{1200} \right) = M \times 1.0025$ $A_2 = A_1 \times 1.0025 + M \times 1.0025$ $= M \times 1.0025^2 + M \times 1.0025$ $= M(1.0025 + 1.0025^2)$	1 mark for solution. Must show A_1 being substituted into A_2.
Q37-ii	$A_n = M(1.0025 + 1.0025^2 + \dots + 1.0025^n)$ $= M \times 1.0025(1.0025 + 1.0025^2 + \dots + 1.0025^{n-1})$ $= M \times 1.0025 \left(\frac{1.0025^n - 1}{1.0025 - 1} \right)$ $= 401M(1.0025^n - 1)$	2 marks for correct working and solution. Must show the geometric progression. 1 mark for showing geometric progression. 1 mark for only using the formula without showing the geometric progression.

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Q37-iii	$S_n = 6000$ $6000 = 401M(1.0025^{12} - 1)$ $M = \frac{6000}{401 \times (1.0025^{12} - 1)}$ $= \$491.93$	2 marks for correct solution 1 mark for equation however did not progress and solve.
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Q37-iv	Amount saved after 5 months $S_n = 401 \times 491.93(1.0025^5 - 1)$ $= \$2478.16$ Amount left to save = $\\$6000 - 2478.16 = \\3521.84 Time required for \$600 payment to reach goal of \$6000. $S_n = 3521.84$ $3521.84 = 401 \times 600(1.0025^n - 1)$ $1.0025^n = \frac{3521.84}{401 \times 600} + 1 = 1.01463$ $n = \frac{\ln(1.01463)}{\ln 1.0025} = 5.81$ therefore 5 months $< n <$ 6 months 5 months total \$600 $S_n = 401 \times 600(1.0025^5 - 1) = \3022.575 5 months compound interest for \$2478.16 $S_5 = 2478.16 \times (1.0025)^5 = \2509.29 Combined total after 10 months = $2509.29 + 3022.58 = 5531.87$ Last month payment = $\\$6000 - \\$5531.87 = \\$468.13 = \\470 (nearest \$5)	3 marks – correct answer 2 marks - Compound interest amount of \$2509.29 - Amount = \$3022.57 - Have found 2 (incorrect) subtotals by considering compound interest and subtracted their 2 subtotals from \$6000 1 mark – - $n = 5.81$
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